THE MARKED MAPPING THEOREM FOR DELOOPABLE TARGETS

Purpose. Informal proofs, all formalised (Section 4): (a) statement M of `doc/delooping-embedding.pdf` is equivalent to its looped form (`MarkLoop`); (b) the restricted 2-torsion lemma, from deloopability alone (`TwoTorsion`); (c) how (b) gives M for every pointed 2-connected 4-type without `Presentation4.Pres4` (`MarkDirect`), hence $\pi_4(S^3) = \mathbb{Z}/2$ using David Wärn's delooping embedding (`Pi4Direct`; not his intended route through a classification of arbitrary maps between triples, see `doc/pi4-s3-embedding.pdf`).

Attribution. The delooping embedding and the sphere strategy are David Wärn's, as is the delooping machinery (`DeloopMaps`); the classification by triples is Hart–Milner's (Milner's equivalence). The cross-term analysis (`CrossTerm`) and the bilinear classification (`H4Bilinear`, Theorem B) are those of `doc/cross-term.pdf` and `doc/h4.pdf`.

Conventions. As in the development, $\mathsf{isConn}\,n$ means $(n-2)$-connected, $\mathsf{isTrunc}\,n$ means $(n-2)$-truncated; $\mathsf{SC4}$ is the type of pointed 2-connected 4-types, $\Omega : \mathsf{SC4} \to \mathsf{SC3}$ looping, $\mathsf{realise}(A, B, k) = \mathrm{Fib}k$ the classification, $\mathsf{Deloopable}\,t = (v : \mathsf{SC4}) \times (\Omega v = \mathsf{realise}\,t)$ (`DeloopEmbedding`). $[X, Y]_*$ is the set truncation of $\mathrm{Map}_*(X, Y)$ (`HomotopyClasses.HC`), $c; d$ composition of classes. $e_m : \Omega K(A, m+1) \simeq_* K(A, m)$ are the Eilenberg–MacLane identifications (`Lev.eA`, `eB`). For $k : K(A,2) \to_* K(B,4)$ put $f_k = e_2; k; e_4^{-1} : \Omega K(A,3) \to_* \Omega K(B,5)$.

Lemma 1 (`SourceS3.LoopEquiv`, $c = 2$). *If* $\mathsf{isConn}\,4\,X$ *and* $\mathsf{isTrunc}\,6\,Y$, *then* $\mathsf{loop} : \mathrm{Map}_*(X, Y) \to \mathrm{Map}_*(\Omega X, \Omega Y)$ *is an equivalence.*

1. Step 1a: the looped form of M

Under the S^3 axiom let $s \in [S^3, BR]_*$ be the class of `SourceS3.Final`, $e = \texttt{omega-BR} : \Omega BR \simeq_* R$, and

$$r = (\mathsf{loop} s); [e] \;\in\; [\Omega S^3, R]_*,$$

where loop on classes is `Sphere3Trunc.loopC` (no representative of s is chosen). For $u = (Y, y, ..) : \mathsf{SC4}$ write $\Omega Y = (y = y)$ and

$$\mathsf{M}(u) :\; s^* : [BR, Y]_* \to [S^3, Y]_* \text{ bijective}; \qquad \mathsf{LM}(u) :\; r^* : [R, \Omega Y]_* \to [\Omega S^3, \Omega Y]_* \text{ bijective}.$$

Lemma 1 (looping on classes)**.** *If* $\mathsf{isConn}\,4\,X$ *and* $\mathsf{isTrunc}\,6\,Y$ *then* $\mathsf{loop} : [X, Y]_* \to [\Omega X, \Omega Y]_*$ *is a bijection.*

Proof. loop on classes agrees on representatives with the map of set truncations induced by loop on pointed maps (`loopC-beta`, `STrMap.beta`; both sides land in a set, so they agree everywhere). The latter is a bijection because loop is an equivalence (Lemma 1; `classes-equiv`). □

Proposition 2 (L)**.** *For every* $u : \mathsf{SC4}$, $\mathsf{M}(u) \iff \mathsf{LM}(u)$.

Proof. Consider the four maps

$$s^* : [BR, Y]_* \to [S^3, Y]_*, \quad \ell = (\mathsf{loop} s)^* : [\Omega BR, \Omega Y]_* \to [\Omega S^3, \Omega Y]_*, \quad \epsilon = [e]^* : [R, \Omega Y]_* \to [\Omega BR, \Omega Y]_*, \quad r^*.$$

(i) $\mathsf{loop}(s; c) = \ell(\mathsf{loop} c)$ (`loopC-comp`). (ii) $r^* c = \ell(\epsilon c)$ (associativity of composition of classes). (iii) ϵ is a bijection (precomposition by a pointed equivalence, `MapFib.pre-pteq`, on classes `pre-equiv`). (iv) $\mathsf{loop} : [BR, Y]_* \to [\Omega BR, \Omega Y]_*$ and $\mathsf{loop} : [S^3, Y]_* \to [\Omega S^3, \Omega Y]_*$ are bijections (Lemma 1: $\mathsf{isConn}\,4\,BR$ is `BR-conn`, $\mathsf{isConn}\,4\,S^3$ is `Sphere3Trunc.Simply.s3-conn`, $\mathsf{isTrunc}\,6\,Y$ is part of u). By (i), (iv) and 2-out-of-3, s^* is bijective iff ℓ is; by (ii), (iii) and 2-out-of-3, r^* is bijective iff ℓ is. □

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Formalised: `MarkLoop.LoopCls` (Lemma 1), `MarkLoop.LMark[S3,base,surf,ump3]`: `r`, `At[u].sq`, `tri`, `e-star`, `to-ls`/`from-ls`, `ls-to-r`/`r-to-ls`, `to-loop`/`from-loop`. Nothing about Y beyond $\mathsf{isTrunc}\,6$ is used, and no presentation.

2. Step 1b: the restricted 2-torsion lemma

Statement 3 (T)**.** Let $t = (A, B, k) : \mathsf{Triples}$ be deloopable, $G : K(\mathbb{Z},2) \to_* K(A,2)$, and let $h : \mathbb{Z} \to B$ with $[G;k] = \Phi_B(h)$ (`H4Proof.h4`; $\Phi_B(h) = [\kappa; K(h,4)]$). Then $2\,h(1) = 0$. In fact $f_{G;k}$ deloops: $\mathsf{Unloop}f_{G;k}$ holds.

2.1. **The proposed reduction to $A = \mathbb{Z}$, checked.** The handoff proposes: deloop G to $\hat{G} : K(\mathbb{Z},3) \to_* \|Y\|_3$ and pull Y back along it. This is correct. (1) $\Omega\|Y\|_3 \simeq \|\Omega Y\|_2 \simeq \|\mathrm{Fib}k\|_2 \simeq K(A,2)$, the last because $\mathsf{fpr} : \mathrm{Fib}k \to K(A,2)$ has 2-connected fibres $k\,x = 0$ (torsors under $\Omega K(B,4)$), so it is the 2-truncation; by construction $\Omega Y \to \Omega\|Y\|_3 \simeq K(A,2)$ is fpr transported along $\Omega Y \simeq_* \mathrm{Fib}k$. (2) $\hat{G}$ exists by Lemma 1 ($K(\mathbb{Z},3)$ is $\mathsf{isConn}\,4$, $\|Y\|_3$ is $\mathsf{isTrunc}\,5$). (3) $Y' = K(\mathbb{Z},3) \times_{\|Y\|_3} Y$ is $\mathsf{isConn}\,4$ (base $\mathsf{isConn}\,4$, fibres those of $Y \to \|Y\|_3$, $\mathsf{isConn}\,5$) and $\mathsf{isTrunc}\,6$. (4) $\Omega Y' \simeq K(\mathbb{Z},2) \times_{K(A,2)} \mathrm{Fib}k \simeq \mathrm{Fib}(G;k)$, loops commuting with pullbacks and (1). So $(\mathbb{Z}, B, G; k)$ is deloopable.

But the reduction does not reduce the difficulty: what remains,

$$\mathrm{Fib}(\kappa; K(h,4)) \text{ deloopable} \implies 2h(1) = 0,$$

is not Lemma N (`Necessity.Nec`), which needs a delooping of the *cocycle* $\kappa; K(h,4)$, i.e. an element of $\mathsf{Unloop}f$ — the Crux of `doc/delooping-embedding.pdf` at $(\mathbb{Z}, B, \kappa; K(h,4))$. A delooping of the *fibre* gives no map $K(\mathbb{Z},3) \to K(B,5)$. The argument below proves the Crux at $A = \mathbb{Z}$ directly, and it handles G by naturality, so the reduction (and its truncation and pullback lemmas) is not needed.

2.2. **Ingredients.** Everything below is stated *pointed in the first variable and pointwise in the second*: no equality of bipointed maps, and no unit coherence in the second variable, is ever needed (review point 1).

The cross term, one variable at a time. For $f : \Omega X \to_* \Omega W$ and $q : \Omega X$ let $\Delta_q^f : \Omega X \to_* \Omega W$,

$$\Delta_q^f(p) = (fp)^{-1} \cdot f(p \cdot q) \cdot (fq)^{-1},$$

with any pointing (by ($\lambda 0$) below only the underlying function matters). The bipointed cross term of `CrossTerm.Cross` is $\delta_f = e_2^{-1}; \mathtt{Dhat}; \tau; \mathtt{Cj}$, and for $x : K(\mathbb{Z},2)$, $q = e_2^{-1}x$, the underlying functions satisfy $\mathrm{ev}_x \circ \delta_f = e_2^{-1}; \Delta_q^f; e_4$ pointwise (`Cross.delta-def`, `parts-def`, `tau-formula`, the definition of `Cj`). Properties, all *pointwise*:

- ($\Delta 0$) $\Delta_q^0(p) = \mathsf{refl}$.
- ($\Delta 1$) For $L = \mathsf{loop}P$, $P : X' \to_* X$, and $p', q' : \Omega X'$: $\Delta_{q'}^{L;f}(p') = \Delta_{Lq'}^f(Lp')$, from $L(p' \cdot q') = Lp' \cdot Lq'$ (`PiOne.Induced.loop-mult`). The same for $L = \hat{G} = e_2; K(a,2); e_2^{-1} = \mathsf{loop}K(a,3)$ (`Lev.loopK`). No pointing is compared, so the unit case ($p' = \mathsf{refl}$) is not needed. (Were it needed, it could not be obtained "by J on q'", a loop with fixed endpoints; one would first prove the unit compatibility of `loop-mult` for free paths and then apply f.)
- ($\delta 2$) (Lemma 13, `Cross.cell-of`, and the cells argument of `DeloopR.unloop-contr`, with $\mathbb{Z}/2$ replaced by B.) For $X = K(\mathbb{Z},3)$, $W = K(B,5)$: if $\delta_f = 0$ then $f = D_p$ for every p, hence $\mathsf{Unloop}f$.

The invariant. For $m : \Omega X \to_* \Omega W$ and $\gamma \in \Omega^3 X$ ($= \Omega^2$ of ΩX) put $\lambda_\gamma(m) = (\mathsf{loop}^2 m)(\gamma) \in \Omega^3 W$. Then

($\lambda 0$) (Lemma: pointing irrelevance) if $m.1 = m'.1$ pointwise, by H, then $\lambda_\gamma(m) = \lambda_\gamma(m')$. Proof, two levels, each conjugation identified before it is removed. (i) Pointwise $(\mathsf{loop}m)(r) = \beta^{-1} \cdot (\mathsf{loop}m')(r) \cdot \beta$ for $r : \Omega X$, with $\beta = m'.2^{-1} \cdot H(\mathsf{refl})^{-1} \cdot m.2 \in \Omega^2 W$ (`loop-ap` on both sides and naturality of H: `LoopConj`). (ii) Since ΩW is a loop space, conjugation by β on $\Omega(\Omega W)$ is pointwise the identity (Eckmann–Hilton), so $\mathsf{loop}m$ and $\mathsf{loop}m'$ have pointwise equal underlying functions, by a homotopy H'. (iii) Applying (i) one level up, $(\mathsf{loop}^2 m)(\gamma) = \beta'^{-1} \cdot (\mathsf{loop}^2 m')(\gamma) \cdot \beta'$ with $\beta' \in \Omega^3 W$, and conjugation on $\Omega^3 W$ is trivial (Eckmann–Hilton). This is the pattern of `Necessity.EPre.At.e-pre` (`LoopConj.LoopHomotopy`, then EH); it is to be stated and proved as its own lemma.

($\lambda 1$) $\lambda_\gamma(L; m) = \lambda_{(\mathsf{loop}^2 L)\gamma}(m)$ (`loop-comp` twice); $\lambda_\gamma(m; e) = (\mathsf{loop}^2 e)(\lambda_\gamma m)$;

($\lambda 2$) $\lambda_\gamma(0) = \mathsf{refl}$ (`loop-const`);

($\lambda 3$) (relation with `EDef.E`, review point 2) with $\gamma_0 = (\mathsf{loop}e_1^{-1})(g) \in \Omega^2 K(\mathbb{Z}, 2)$ the generator (g the generator of the circle) and $\hat{\gamma}_0 = (\mathsf{loop}^2 e_2^{-1})(\gamma_0) \in \Omega^3 K(\mathbb{Z}, 3)$, for every bipointed c and x:

$$(E\, c)(x) = (\mathsf{loop}^2(\mathrm{ev}_x \circ c))(\gamma_0), \qquad \text{hence} \qquad (E\, \delta_f)(x) = (\mathsf{loop}^2 e_4)\big(\lambda_{\hat{\gamma}_0}(\Delta^f_{e_2^{-1}x})\big).$$

To be stated and proved as its own lemma. The first equation is `ENat.E-loop` and `L4.ev-ap` (ev_x commutes with looping), with $\mathsf{loop}(\mathtt{phi}\, c; \mathtt{Ex}\, x)$ at g rewritten by `loop-comp`. For the second, compose with e_4^{-1} so that both sides are pointed maps into $\Omega K(B, 5)$ with the same underlying function ($\mathrm{ev}_x \circ \delta_f$ above), apply ($\lambda 0$) there (this is where the conjugations are identified and removed), and use ($\lambda 1$) for e_2^{-1} and e_4. Also $(\mathsf{loop}^2 e_4)(\mathsf{refl}) = \mathsf{refl}$ by the pointing lemma of loop (`loop-const` style).

($\lambda 4$) (Theorem B) $\Theta c = E\, c; j_B : K(\mathbb{Z}, 2) \to_* K(B, 2)$ and `BilAt.theta-inj`: $\Theta c = \Theta c'$ implies $c = c'$. The type $\mathrm{Map}_*(K(\mathbb{Z}, 2), K(B, 2))$ is a set (`cor5-map`: $K(\mathbb{Z}, 2)$ is 1-connected, $K(B, 2)$ a 2-type), and $\Omega K(B, 2)$ is connected, so equality there follows from equality of the underlying functions (pointing irrelevance, `H4Classes.pi-set`).

Two facts about $\mathsf{fpr} : \mathrm{Fib}k \to K(A, 2)$, transported along the pointed equivalences ϵ, e_2 to π below.

(F1) $\Omega^2 \mathsf{fpr} : \Omega^2 \mathrm{Fib}k \to \Omega^2 K(A, 2)$ is surjective. By `Fibres.omega-fib` twice, $\Omega^2 \mathrm{Fib}k \simeq_* \mathrm{Fib}(\mathsf{loop}^2 k)$ over $\Omega^2 K(A, 2)$; the fibre over x is $(\mathsf{loop}^2 k)(x) = \mathsf{refl}$, an identity type of $\Omega^2 K(B, 4) \simeq K(B, 2)$, which is 1-connected; so the fibre is connected, in particular merely inhabited.

(F2) (review point 4: stated for *functions*, unpointed) For a family of 2-types T over $K(A, 2)$, precomposition with fpr is an equivalence $\big((x : K(A, 2)) \to T\, x\big) \simeq \big((w : \mathrm{Fib}k) \to T(\mathsf{fpr}\, w)\big)$, since the fibres of fpr are 2-connected (`ConnectedMaps.Elim`). In particular a function $\varphi : K(A, 2) \to T$ into a 2-type with $\varphi(\mathsf{fpr}\, w) = t_0$ for all w satisfies $\varphi(x) = t_0$ for all x (take the family $\varphi(x) = t_0$, a family of 1-types).

2.3. **The proof.** Let $\epsilon : \Omega Y \simeq_* \mathrm{Fib}k$ witness $\mathsf{Deloopable}\, t$ (`PtProp` for SC3), $\pi = \epsilon; \mathsf{fpr}; e_2^{-1} : \Omega Y \to_* \Omega K(A, 3)$, $f' = f_{G;k}$.

Step 1 (π is a loop map). By Lemma 1 ($\mathsf{isConn}\, 4\, Y$; $K(A, 3)$ is $\mathsf{isTrunc}\, 5 \subseteq \mathsf{isTrunc}\, 6$), $\pi = \mathsf{loop}P$ for some $P : Y \to_* K(A, 3)$.

Step 2 ($\pi; f_k$ is null). $\pi; f_k = \epsilon; \mathsf{fpr}; e_2^{-1}; e_2; k; e_4^{-1} = \epsilon; (\mathsf{fpr}; k); e_4^{-1}$, and $\mathsf{fpr}; k$ is merely 0 (the analogue of `MapFib.SourceP.fseq` for any k, to be written: $(x, w) \mapsto w$ and pointing irrelevance). So merely $N : \pi; f_k = 0$ in $\mathrm{Map}_*(\Omega Y, \Omega K(B, 5))$.

Step 3 (transfer along G). $G = K(a, 2)$ (`H2`, Lemma 16), and $f' = \hat{G}; f_k$ with $\hat{G} = e_2; G; e_2^{-1} = \mathsf{loop}K(a, 3)$ (`loopK`).

Step 4 (the goal is a proposition, review point 3). We prove $\Theta\,\delta_{f'} = \Theta\,0$, an equality in the set $\mathrm{Map}_*(K(\mathbb{Z},2), K(B,2))$ ($\lambda 4$). It is a proposition, so we may eliminate into it the mere path N of Step 2 and a mere $\gamma' \in \Omega^3 Y$ with $(\mathsf{loop}^2\pi)(\gamma') = (\mathsf{loop}^2\hat{G})(\hat{\gamma}_0)$, which exists by (F1).

Step 5 (the invariant vanishes). Define $\varphi : \Omega K(A,3) \to \Omega^3 K(B,5)$, $\varphi(y) = \lambda_{\gamma'}(\pi; \Delta_y^{f_k})$. For $b : \Omega Y$,

$$\varphi(\pi b) = \lambda_{\gamma'}(\Delta_b^{\pi; f_k}) = \lambda_{\gamma'}(\Delta_b^0) = \lambda_{\gamma'}(0) = \mathsf{refl}$$

by ($\lambda 0$) with ($\Delta 1$) at $L = \pi = \mathsf{loop}P$, ap of $g \mapsto \lambda_{\gamma'}(\Delta_b^g)$ along N, ($\lambda 0$) with ($\Delta 0$), ($\lambda 2$). *This is the only use of deloopability*: the multiplicativity of π. The target $\Omega^3 K(B,5) \simeq K(B,2)$ is a 2-type, so by (F2) (along ϵ, e_2) $\varphi(y) = \mathsf{refl}$ for all y. Now for $x : K(\mathbb{Z},2)$, $q = e_2^{-1}x$:

$$\begin{aligned}
\lambda_{\hat{\gamma}_0}(\Delta_q^{f'}) &= \lambda_{\hat{\gamma}_0}(\hat{G}; \Delta^{f_k}_{\hat{G}q}) = \lambda_{(\mathsf{loop}^2\hat{G})\hat{\gamma}_0}(\Delta^{f_k}_{\hat{G}q}) && (\lambda 0),\ (\Delta 1) \text{ at } L = \hat{G},\ (\lambda 1)\\
&= \lambda_{(\mathsf{loop}^2\pi)\gamma'}(\Delta^{f_k}_{\hat{G}q}) = \lambda_{\gamma'}(\pi; \Delta^{f_k}_{\hat{G}q}) = \varphi(\hat{G}q) = \mathsf{refl} && \text{choice of } \gamma',\ (\lambda 1).
\end{aligned}$$

By ($\lambda 3$), $(E\,\delta_{f'})(x) = (\mathsf{loop}^2 e_4)(\mathsf{refl}) = \mathsf{refl} = (E\,0)(x)$ (`ETwo.e-zero`), so $\Theta\,\delta_{f'}$ and $\Theta\,0$ have equal underlying functions, hence are equal ($\lambda 4$). By `theta-inj`, $\delta_{f'} = 0$. (These last two steps are the body of `Necessity.ThetaEq.theta-eq`; it cannot be applied as it stands, since its hypothesis, the pointwise equality of E, is not a proposition and so cannot receive the mere N and γ'. We apply its components, `pi-set` with `BL.sK2`, `BL.cK2`, and `theta-inj`, after the elimination; or split `theta-eq` at `Theta`.)

Step 6. By ($\delta 2$), $\mathsf{Unloop}f'$. The goal $2h(1) = 0$ is a proposition and $G; k = \kappa; K(h,4)$ merely (effectiveness of $[G;k] = \Phi_B(h)$), so we may transport to $\mathsf{Unloop}f_{\kappa;K(h,4)}$, and Corollary conv (`Necessity.Conv`, i.e. Lemma N) gives $2h(1) = 0$. (The orientations agree: $f_{\kappa;K(h,4)}$ is `Conv`'s `CJ.conj F`, conjugation by e_2 of $\mathbb{Z}$ and e_4 of B; `Cross` uses the locked `Lev.eB`, equal to `KK.e` by `eA-def`/`eB-def`.) □

Remark 4. Taking $G = K(\mathrm{id},2)$ at $A = \mathbb{Z}$, Steps 1–6 prove the Crux of `doc/delooping-embedding.pdf` at every triple $(\mathbb{Z}, B, k)$: if Fib$k$ deloops, so does f_k. For general A the same argument would need Theorem B with source $K(A,2)$, which the development does not have; Statement T only needs sources $K(\mathbb{Z},2)$.

Remark 5 (classically). Steps 3–5 say: the cross term of k, a class in $\tilde{H}^4(K(A,2) \wedge K(A,2); B)$, vanishes after pulling back to Fib$k \wedge$ Fib$k$, because Fib$k$ is an H-space over $K(A,2)$; and pulling back is injective on the $\pi_2 \otimes \pi_2$ part, which is what E detects. The cross term of $\kappa; K(h,4)$ is $2h\,\iota \otimes \iota$; we do not need this computation, since Lemma N is used instead.

3. Step 1c: M for deloopable targets, and compatibility with r

Let $u = (Y, ..) : \mathsf{SC4}$ and $t = (A, B, k)$ with $\Omega Y \simeq_* \mathrm{Fib}k$ (`sc4-equiv`: every u arises so). By Proposition 2 and transport along $\Omega Y \simeq_* \mathrm{Fib}k$ (`ptEqInd`, as `rep-pteq`), M at u follows from

$$r^* : [R, \mathrm{Fib}k]_* \to [\Omega S^3, \mathrm{Fib}k]_* \text{ bijective.}$$

Compose with $q = \mathsf{fpr}_*$ on both sides: $q(r;c) = r; q(c)$ (associativity), so the square commutes. Then:

(1) $r^* : [R, K(A,2)]_* \to [\Omega S^3, K(A,2)]_*$ is bijective: Proposition L at $u = K(A,3) : \mathsf{SC4}$, where M holds (`SourceS3.Final.right`), then e_2.

(2) q is bijective from the source R (`MapFib.Lifting`-type lemma for k, with $\Omega K(B,4)$ and ΩFib$k$ connected): (E) every $c : R \to_* K(A,2)$ is $p_R; G$ on classes (p_R^* bijective into $K(A,2)$ for $R = \mathrm{Fib}\bar{k}$: a new lemma, the proof of `MapFib.SourceP.PStar` with κ replaced by $\bar{k}$, the fibres of p_R being 2-connected), and $[p_R; G; k] = [p_R; \bar{k}; K(g,4)] = 0$ because $2h(1) = 0$ (Statement T), $h = \rho; g$ (`Factor`), $\Phi(\rho; g) = [\bar{k}; K(g,4)]$ (`Psi`), and $p_R; \bar{k}$ is merely null (the generic `fseq` of Step 2); (U) $[R, K(B,3)]_* = 0$ (`MapBR.UR`, `FibreH3.FibAtC` at $(\mathbb{Z}/2, \bar{k})$).

(3) q is bijective from the source ΩS^3: (E) by 1, every class is $r;c$ with c from R, and $[r;c;k] = r;[c;k] = 0$ by 2(E); (U) $[\Omega S^3, \Omega K(B,4)]_* \cong [S^3, K(B,4)]_* = 0$ (Lemma 1 at $K(B,4)$, `s3-paths`: $\Omega^3 K(B,4)$ is connected).

By 2-out-of-3, r^* is bijective into Fib$k$. This proves M for every $u : \mathsf{SC4}$ with no presentation, and the only new input is Statement T.

4. Formalisation: status and correspondence

Everything above is formalised and checked (`./segc sag/pi3-s2-postnikov all`: 96 modules, 0 failed; the 26 `.opens` specifications, rule 6 of `seg/WRITING.md`, have no deviation). The correspondence, text → declaration (modules of `sag/pi3-s2-postnikov`):

Lemma 1, Proposition 2	`MarkLoop.LoopCls`, `MarkLoop.LMark.r`, `LMark.At.to-loop/from-loop`
(λ0)	`CrossDeloop.PtIrr.lambda0` (`LoopIrr`, `conj-eh`)
(λ1), (λ2)	`CrossDeloop.lam-pre`, `Inv.lam-zero`
(λ3)	`CrossDeloop.ELoop.At.e-loop`, `EDelta.At.e-delta`
(λ4)	`H4Bilinear.BilAt.theta-inj`, `Necessity.ThetaEq.theta-pt`
Δ_q^f, (Δ0), (Δ1)	`CrossDeloop.Inv.Delta`, `Inv.delta0`, `Delta1.d1`
δ_f, (δ2)	`CrossTerm.CrossGen[B,f].delta`, `cell-of`; `DeloopR.UnloopGen.unloop-contr`
(F1), (F2), generic `fseq`	`MapFib.FibP.surj2`, `FibP.const-ext`, `FibP.fseq`
Step 1	`TwoTorsion.Loop.eps`, `pi`, `pi-cell`, `pi-loop`, `pi-mult`
Step 2	`TwoTorsion.Loop.N` (from `FibP.fseq`, eliminated in Step 4)
Step 3	`TwoTorsion.TwoTors.Ghat`, `ghat-loop`, `ghat-mult`, `f'-eq`
Step 4	`TwoTors.tgt`, `xA`; the module `TwoTors.Elim[n,w,ew]` receives the mere N and γ'; `TwoTors.theta` eliminates them into $\Theta\,\delta = \Theta\,0$ (`BL.sK2`)
Step 5	`Elim.gamma'`, `eg`, `phi`, `phi-pi`, `phi-refl`, `lam-fp`, `h-cell`, `e-zero`, `theta`
Step 6	`TwoTors.cross`, `unloop` (Unloop$f_{G;k}$), `TwoTors.Two[h,eh].two-tors` ($2h(1) = 0$; `conj-bridge`, `Necessity.Conv`)
Section 3, 2(E), 2(U)	`MarkDirect.ExR` (with `MapFib.FibP[Z,Z/2,`$\bar{k}$`].PStar`), `ex-R`, `un-R` (`MapBR.SourceBR.ur`), `qR`
Section 3, item 1	`MarkDirect.Direct.uA`, `markedA` (`SourceS3.Final.right`), `rA`, `transfer`
Section 3, item 3	`Direct.ExS`, `UnS`, `qS`
r^* bijective into Fib$k$; M for every u	`Direct.rF`; `Direct.AtU[u].marked`, `Direct.marked`
$\pi_4(S^3) = \mathbb{Z}/2$	`Pi4Direct.Pi4S3.pi4-S3` (`Pi4Sphere.Two4`, `Pi4Sphere.pi4-BR`)

Dependencies. Statement T (`TwoTorsion`) has no sphere parameter and uses no presentation; its only use of deloopability is `Elim.phi-pi`. `MarkDirect` uses Statement T and the S^3 axiom (through r, `SourceS3.Final.right` and `s3-paths`). The marked class is the existing one: r = `MarkLoop.LMark.r` and s = `SourceS3.Final.s`. The new endpoint `Pi4Direct.Pi4S3.pi4-S3` does not depend on `Presentation4.Pres4` (nor `Final4`) nor on `DeloopEmbedding.Mark.marked`: a copy of the development with `Presentation4.seg` deleted,

`Mark.At`/`Mark.marked` and `Pi4Sphere.Assemble4`/`Pi4S3` removed still checks (4874 declarations, 0 failed). `Pi4Sphere.Pi4S3` (through `Pres4`) is kept unchanged as the reference.

Changes to existing modules. `MapFib.SourceP.PStar` became `MapFib.FibP.PStar` (Lemma 7 for any triple; `SourceP` uses it at $(\mathbb{Z},\mathbb{Z},\kappa)$); `Necessity.ThetaEq.theta-eq` was split at Θ (`theta-pt`), as anticipated in Step 5. Two locks were added for rule 6: `Loop.eps` (only its type is used) and `Direct.AtU.t` (after its equation `t-eq`); without them a by-definition step opened their constructions.

Review. A first review (28 Oct) asked for: (Δ1) only for actual loop maps, with the pointings explicit; the raw and conjugated conventions kept apart ($\hat{\gamma}_0$); the propositional eliminations explicit (Step 4); (F2) at the level of maps, not classes. The present text follows these: the argument is pointwise in both variables, the pointings being removed once, by (λ0), and it eliminates into the set-level equality $\Theta\,\delta = \Theta\,0$. A second, independent review checked the revised text against the code. Its corrections are included: (λ0) replaces an incorrect "by J on q'" for the pointing in (Δ1); the two-level Eckmann–Hilton step is made explicit; the citations for `fseq`, p_R^* and Corollary C are corrected. A third reading asked that (λ0) and (λ3) be separate lemmas with the conjugations identified (`LoopConj`) before Eckmann–Hilton removes them; they were formalised first (`CrossDeloop`).